Directional Derivatives, Chain Rule, and Optimization
Lecture 4 for Multivariable Calculus

Contents

1 Lecture Scope and Goals

In this lecture, we build directly on Lecture 3. The main goals are:

- define directional derivatives and connect them to the gradient,

- apply chain rule in one-parameter and multi-parameter settings,
- interpret the Jacobian matrix as the best linear map,
- solve unconstrained optimization problems in two variables,
- solve constrained optimization problems using Lagrange multipliers.

Central Theme

Partial derivatives describe change along coordinate axes. Directional derivatives and the chain rule tell us how functions change along *any* motion.

2 Directional Derivatives and Gradient

Let $f : \mathbb{R}^2 \to \mathbb{R}$, point $a = (a, b)$, and direction vector $u = (u_1, u_2)$ with $\|u\| = 1$.

2.1 Definition

The directional derivative of f at a in direction u is

$$D_u f(a) = \lim_{t\to 0} \frac{f(a + tu) - f(a)}{t},$$

if the limit exists.

2.2 Key Formula (Differentiable Case)

If f is differentiable at a, then

$$D_u f(a) = \nabla f(a) \cdot u.$$

So the directional derivative is the projection of $\nabla f(a)$ onto direction u.

2.3 Geometric Meaning

- $\nabla f(a)$ points in the direction of steepest increase.
- The maximum directional derivative is $\|\nabla f(a)\|$, achieved at $u = \dfrac{\nabla f(a)}{\|\nabla f(a)\|}$.
- The minimum directional derivative is $-\|\nabla f(a)\|$.
- If $u \perp \nabla f(a)$, then $D_u f(a) = 0$ (instantaneously no change).

Example 1. Let

$$f(x, y) = x^2y + y^3.$$

Then

$$\nabla f = (2xy,\, x^2 + 3y^2).$$

At $(1, 1)$,

$$\nabla f(1, 1) = (2, 4).$$

For direction $u = \left(\frac{3}{5}, \frac{4}{5}\right)$,

$$D_u f(1,1) = \nabla f(1,1) \cdot u = (2,4) \cdot \left(\frac{3}{5}, \frac{4}{5}\right) = \frac{22}{5}.$$

Pitfall 1

The direction vector must be a *unit* vector in the formula $D_u f = \nabla f \cdot u$. If a non-unit vector v is used, normalize first: $u = \dfrac{v}{\|v\|}$.

3 Tangent Plane and Normal Line (Applied View)

For $z = f(x,y)$ differentiable at (a,b):

$$z \approx f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b).$$

This is the tangent plane approximation.

3.1 Normal Vector to Tangent Plane

For surface $F(x,y,z) = f(x,y) - z = 0$, a normal vector is

$$\nabla F(a,b,f(a,b)) = (f_x(a,b), f_y(a,b), -1).$$

Any nonzero scalar multiple is also normal, so both n and $-n$ are valid normals.

3.2 How to Choose the Normal *Direction*

A surface has two opposite normal directions at a smooth point. In applications we often need a consistent choice (an *orientation*).

1) Implicit surface $F(x,y,z) = c$. The gradient ∇F points toward increasing values of F. So:

- $+\nabla F$ and $-\nabla F$ are both normal directions,
- choose the sign using words in the problem: *outward*, *upward*, *inward*, etc.

2) Graph surface $z = f(x,y)$. For $F(x,y,z) = f(x,y) - z$, a normal is $(f_x, f_y, -1)$; the opposite normal is $(-f_x, -f_y, 1)$. The second one has positive z-component, so it is the *upward* normal.

3) Parametric surface $\mathbf{r}(u,v)$. Use

$$n = \mathbf{r}_u \times \mathbf{r}_v.$$

Reversing parameter order flips direction:

$$\mathbf{r}_v \times \mathbf{r}_u = -(\mathbf{r}_u \times \mathbf{r}_v).$$

So orientation depends on parameter order.

Example (Sphere). For sphere $x^2 + y^2 + z^2 = R^2$, take

$$F(x, y, z) = x^2 + y^2 + z^2, \quad \nabla F = (2x, 2y, 2z).$$

At point $p = (x_0, y_0, z_0)$, $\nabla F(p)$ points away from the origin, so it is the *outward* normal direction. The inward normal is $-\nabla F(p)$.

Example 2. For $f(x, y) = x^2 + xy + y^2$ at $(1, 2)$:

$$f(1, 2) = 7, \quad f_x = 2x + y, \quad f_y = x + 2y,$$

so

$$f_x(1, 2) = 4, \quad f_y(1, 2) = 5.$$

Tangent plane:

$$z = 7 + 4(x - 1) + 5(y - 2).$$

A normal vector is $(4, 5, -1)$.

3.3 Example 2 via Level Curve (Mathematica View)

For

$$f(x, y) = x^2 + xy + y^2, \qquad f(1, 2) = 7,$$

the level curve through $(1, 2)$ is

$$x^2 + xy + y^2 = 7.$$

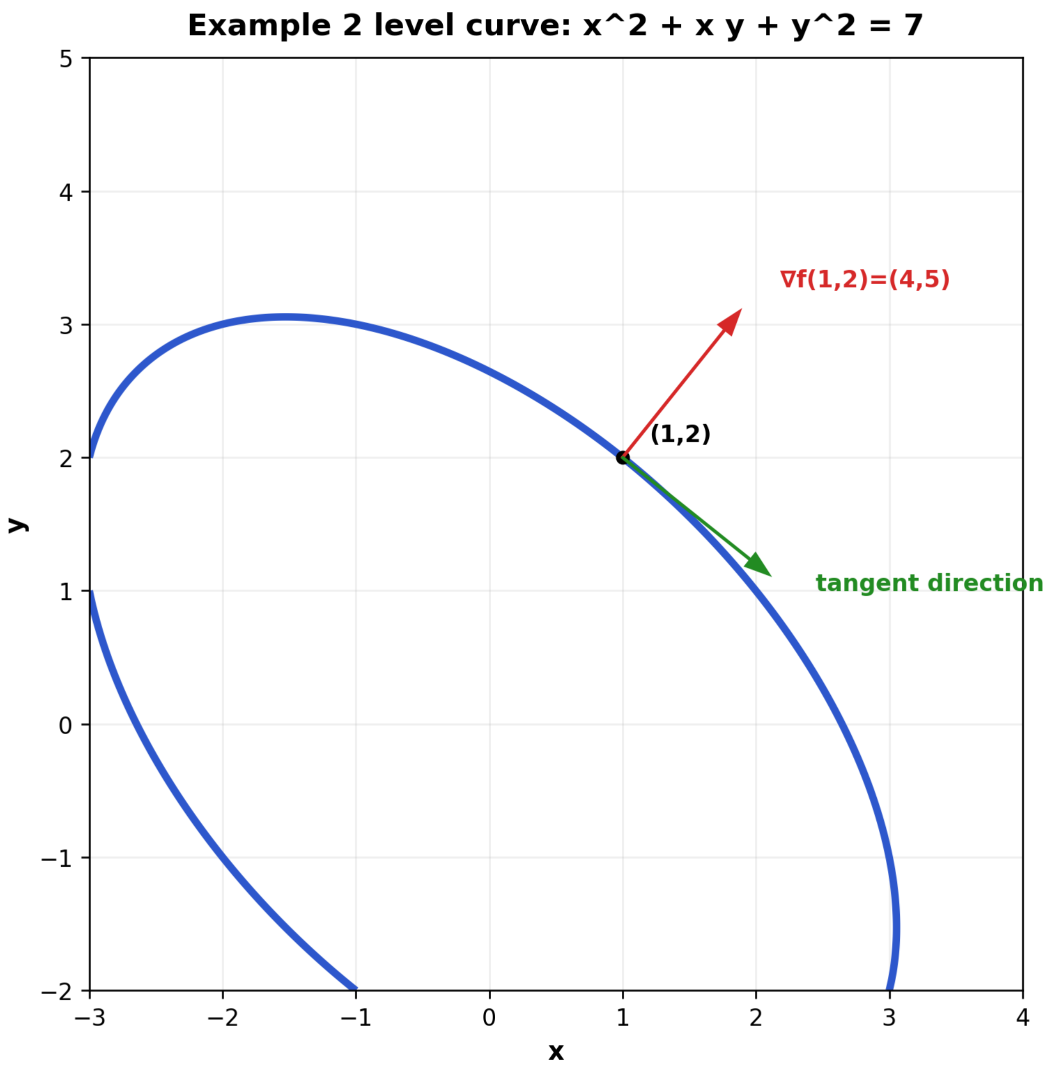

Figure 1: Level curve $f(x, y) = 7$ with the gradient vector and a tangent direction at $(1, 2)$.

At $(1, 2)$,

$$\nabla f(1, 2) = \big(f_x(1, 2), f_y(1, 2)\big) = (4, 5).$$

For any tangent vector t to the level curve, f is constant along the curve, so

$$\frac{\mathrm{d}}{\mathrm{d}s} f(\gamma(s)) = \nabla f \cdot \gamma'(s) = \nabla f \cdot t = 0.$$

Hence $\nabla f(1, 2)$ is normal to the level curve in the xy-plane. For the graph surface $F(x, y, z) = f(x, y) - z = 0$, this becomes the 3D normal

$$\nabla F(1, 2, 7) = (4, 5, -1),$$

which is exactly the normal used in the tangent-plane equation above.

3.4 Visualizing Normal Direction in 3D

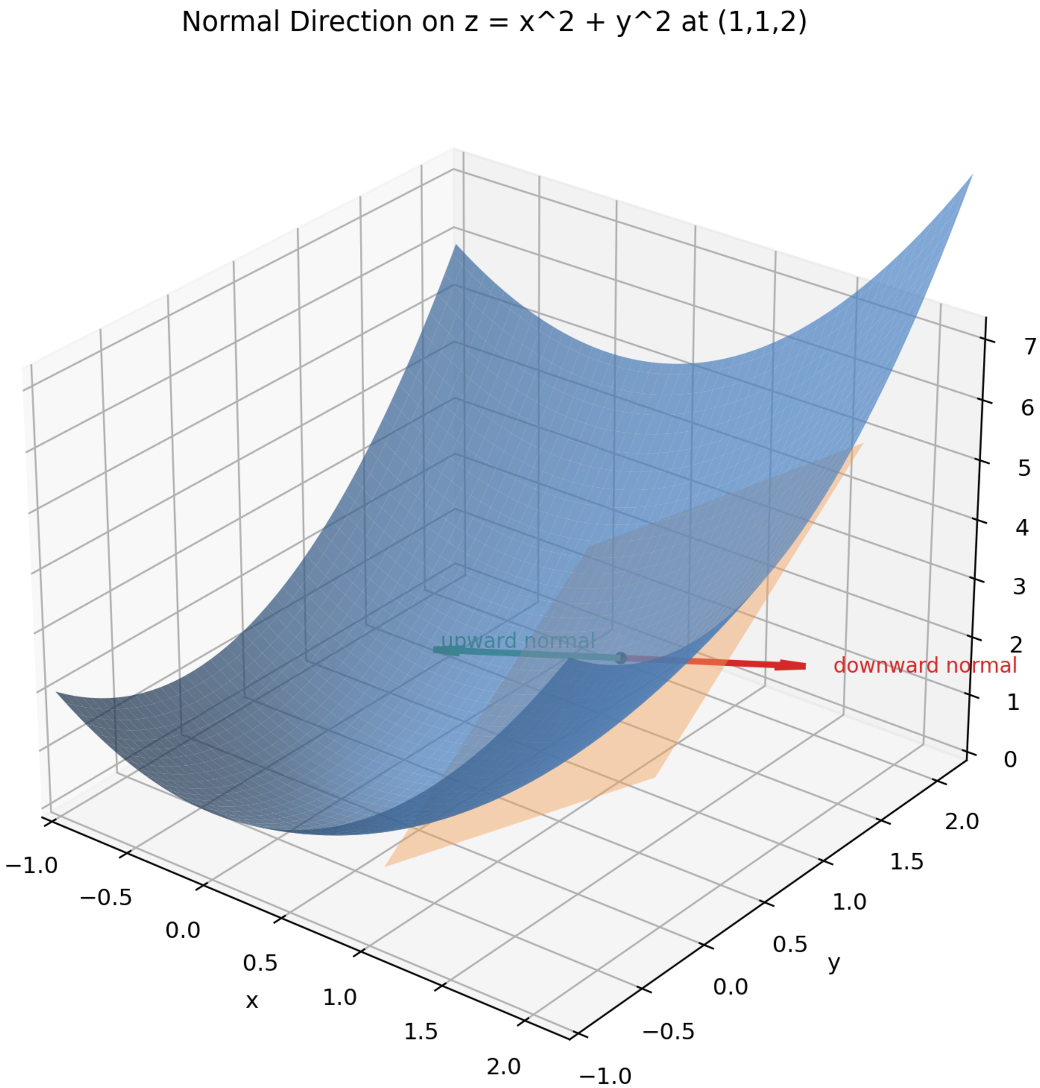

Figure 2: For graph surface $z = x^2+y^2$, two opposite normal directions at the same point correspond to opposite choices of orientation.

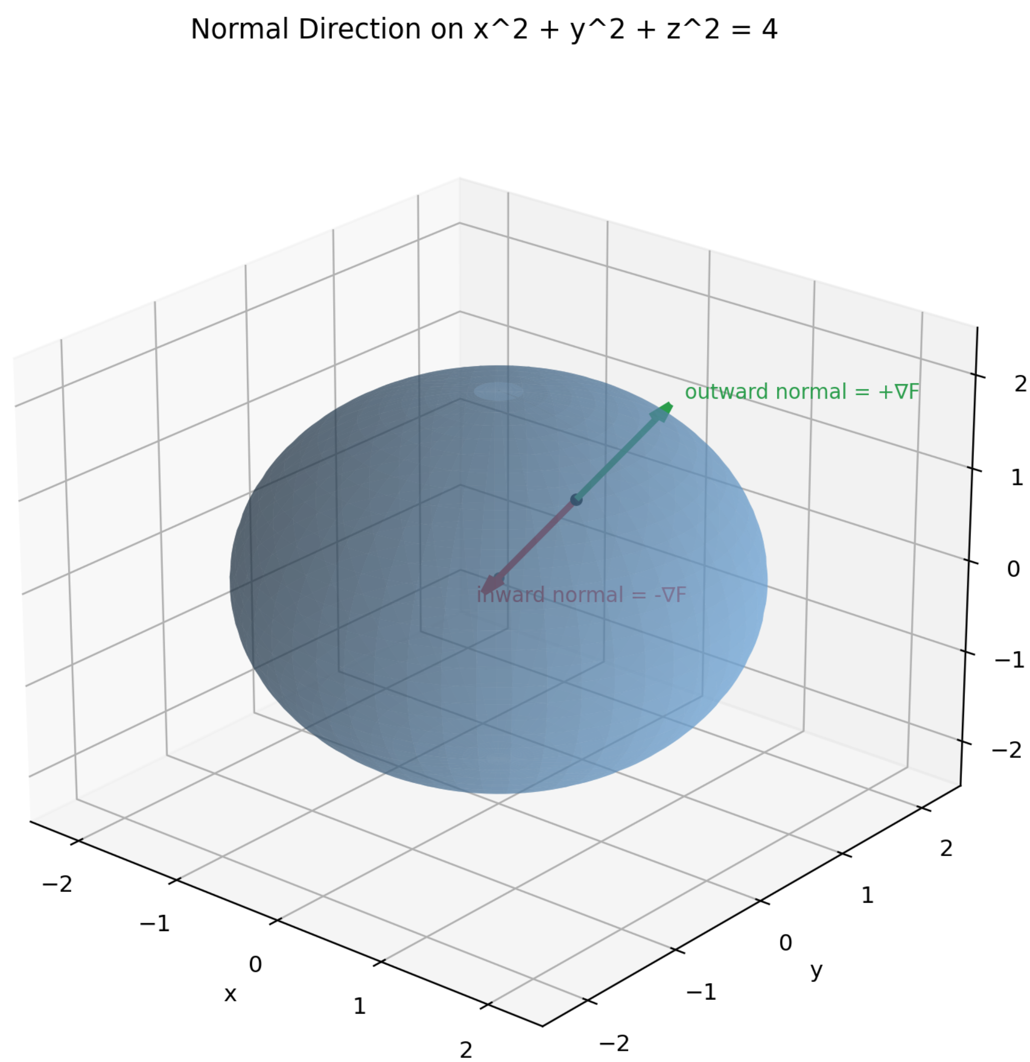

Figure 3: For sphere $x^2 + y^2 + z^2 = 4$, ∇F gives the outward normal direction and $-\nabla F$ gives the inward normal direction.

4 Chain Rule in Multivariable Calculus

4.1 Why Gradient Gives Normal Direction (Geometric Meaning)

Let a level surface be

$$F(x, y, z) = c.$$

Take any smooth curve $\mathbf{r}(t)$ lying on this surface. Then $F(\mathbf{r}(t)) = c$, so

$$\frac{\mathrm{d}}{\mathrm{d}t} F(\mathbf{r}(t)) = 0.$$

By the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t} F(\mathbf{r}(t)) = \nabla F(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0.$$

Here $\mathbf{r}'(t)$ is a tangent vector to the surface, so ∇F is orthogonal to every tangent direction. Therefore ∇F is a normal vector of the level surface.

Geometric meaning.

- ∇F points in the direction of fastest increase of F.
- $\|\nabla F\|$ is the maximal rate of increase per unit distance.
- Along tangent directions, first-order change is zero, so tangent directions are perpendicular to ∇F.

4.2 Case A: One Parameter t

Let $x = x(t),\ y = y(t)$ and $z = f(x, y)$. Then

$$\frac{\mathrm{d}z}{\mathrm{d}t} = f_x \frac{\mathrm{d}x}{\mathrm{d}t} + f_y \frac{\mathrm{d}y}{\mathrm{d}t} = \nabla f \cdot (x'(t), y'(t)).$$

Example 3. Let $f(x, y) = x^2 y$, with $x = t^2$, $y = e^t$.

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x(t), y(t)) = f_x x' + f_y y'.$$

Compute:

$$f_x = 2xy, \quad f_y = x^2, \quad x' = 2t, \quad y' = e^t.$$

So

$$\frac{\mathrm{d}z}{\mathrm{d}t} = 2xy(2t) + x^2 e^t = 4txy + x^2 e^t.$$

Substitute $x = t^2,\ y = e^t$:

$$\frac{\mathrm{d}z}{\mathrm{d}t} = 4t(t^2)e^t + t^4 e^t = (4t^3 + t^4)e^t.$$

4.3 Case B: Two Parameters (s,t)

If $x = x(s, t),\ y = y(s, t)$ and $w = f(x, y)$, then

$$\frac{\partial w}{\partial s} = f_x \frac{\partial x}{\partial s} + f_y \frac{\partial y}{\partial s}, \qquad \frac{\partial w}{\partial t} = f_x \frac{\partial x}{\partial t} + f_y \frac{\partial y}{\partial t}.$$

Example 4. Let $f(x, y) = e^{xy}$, $x = s + t$, $y = s - t$. Then

$$xy = (s + t)(s - t) = s^2 - t^2,$$

so

$$w(s, t) = e^{s^2 - t^2}.$$

Directly,

$$\frac{\partial w}{\partial s} = 2se^{s^2 - t^2}, \qquad \frac{\partial w}{\partial t} = -2te^{s^2 - t^2}.$$

These match chain rule computations.

5 Jacobian Matrix and Linearization

For $F : \mathbb{R}^n \to \mathbb{R}^m$ with components $F = (F_1, \dots, F_m)$, the Jacobian at a is

$$J_F(a) = \left[\frac{\partial F_i}{\partial x_j}(a) \right]_{m \times n}.$$

5.1 Interpretation

If F is differentiable at a,

$$F(a+h) = F(a) + J_F(a)h + o(\|h\|).$$

So $J_F(a)$ is the best linear approximation.

Example 5. Let

$$F(x, y) = \big(x^2 - y,\, xy\big).$$

Then

$$J_F(x, y) = \begin{bmatrix} 2x & -1 \\ y & x \end{bmatrix}.$$

At $(1, 2)$:

$$J_F(1, 2) = \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix}.$$

6 Unconstrained Optimization in Two Variables

6.1 Critical Points

For $f : \mathbb{R}^2 \to \mathbb{R}$, a point (a, b) is critical if

$$\nabla f(a, b) = (0, 0)$$

or one/both first partials do not exist.

6.2 Second Derivative Test

Let

$$A = f_{xx}(a, b), \quad B = f_{xy}(a, b), \quad C = f_{yy}(a, b), \quad D = AC - B^2.$$

Then:

- If $D > 0$ and $A > 0$: local minimum.
- If $D > 0$ and $A < 0$: local maximum.
- If $D < 0$: saddle point.
- If $D = 0$: inconclusive.

Example 6.

$$f(x, y) = x^3 - 3x + y^2.$$

First derivatives:

$$f_x = 3x^2 - 3, \quad f_y = 2y.$$

Critical points: $x = \pm 1$, $y = 0$, so $(1, 0)$ and $(-1, 0)$. Second derivatives:

$$f_{xx} = 6x, \quad f_{xy} = 0, \quad f_{yy} = 2.$$

At $(1, 0)$:

$$D = (6)(2) - 0 = 12 > 0, \ A = 6 > 0 \Rightarrow \text{local minimum.}$$

At $(-1, 0)$:

$$D = (-6)(2) - 0 = -12 < 0 \Rightarrow \text{saddle point.}$$

Pitfall 2

Second derivative test classifies only local behavior near critical points. It does not automatically give a global maximum/minimum on unbounded domains.

7 Constrained Optimization: Lagrange Multipliers

To optimize $f(x, y)$ subject to $g(x, y) = c$, solve

$$\nabla f(x, y) = \lambda \nabla g(x, y), \qquad g(x, y) = c.$$

7.1 Why It Works (Geometric Intuition)

At an optimal point on a smooth constraint curve, the level curve of f is tangent to the constraint. Their normals are parallel, so gradients are parallel.

Example 7. Find max/min of

$$f(x, y) = xy$$

subject to

$$x^2 + y^2 = 1.$$

Set

$$\nabla f = (y, x), \quad \nabla g = (2x, 2y),$$

so

$$(y, x) = \lambda(2x, 2y).$$

System:

$$y = 2\lambda x, \qquad x = 2\lambda y, \qquad x^2 + y^2 = 1.$$

From first two equations, $x = \pm y$. On unit circle:

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

Function values:

$$f = \frac{1}{2}, \ \frac{1}{2}, \ -\frac{1}{2}, \ -\frac{1}{2}.$$

So max is $\frac{1}{2}$, min is $-\frac{1}{2}$.

8 Mini Workflow for Students

Use this checklist for Lecture 4 problems:

1. If asked for rate of change in direction, normalize direction vector first.
2. For surface normals, first compute a valid normal, then pick sign to match orientation words (*upward/outward/inward*).
3. Compute gradient, then use $D_u f = \nabla f \cdot u$.
4. For composite variables, write dependence graph and apply chain rule carefully.
5. For extrema without constraints: find critical points and use second derivative test.
6. For constraints: solve $\nabla f = \lambda \nabla g$ together with the constraint equation.

9 Practice Set (In-Class or Homework)

1. Let $f(x, y) = x^2 + 2xy - y^2$. Compute $D_u f(1, -1)$ for $u = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.
2. Let $z = x^2 + y^2$, where $x = t \cos t$, $y = t \sin t$. Find $\frac{\mathrm{d}z}{\mathrm{d}t}$.
3. Find and classify critical points of $f(x, y) = x^2 + y^2 - 4x + 6y$.
4. Use Lagrange multipliers to find extrema of $f(x, y) = x + 2y$ on $x^2 + y^2 = 5$.
5. Let $F(x, y) = (x^2 + y,\, e^{xy})$. Compute $J_F(0, 1)$ and linear approximation near $(0, 1)$.

Takeaway

Lecture 4 extends derivative ideas from local linearization to motion, direction, and optimization. The gradient unifies these topics: it controls directional change, powers chain rule structure, and identifies extrema with or without constraints.